

Session Topic: Game Theory (Chapter 9) → Monopolistic Competition (Chapter 14)

Learning objectives (Game Theory)

- Apply game theory as a mathematical framework for analyzing situations of interdependence — where one party's payoff depends on another's choices.
- Learn the basic definitions and solution concepts used to analyze games.
- See how game theory explains real-world problems, including illicit drug markets and the economics of climate change, as well as pricing behavior in oligopoly and monopolistic competition.
- Distinguish rational, self-interested behavior from models of cooperation and non-self-interested (non-rational) behavior.

Core game theory concepts covered

- Best response — a player's optimal move given what the other player is expected to do.
- Nash equilibrium — a set of strategies where no player can benefit by changing their own strategy alone.
- Dominant strategy — a strategy that is best regardless of what the other player does.
- Worked textbook examples referenced during class: a betting/mining-themed game, a comparative-advantage game, and the classic prisoner's dilemma (tied to specific numbered tables in Ch. 9 — exact table numbers were unclear in the recording, likely in the Table 9.2–9.4 range).
- A public-goods game: two neighbors decide whether to each pay an equal share toward a shared community benefit. Used to illustrate the free-rider problem — individuals may prefer to enjoy a shared good without paying for it, even though everyone is better off if it gets funded.

Transition to Monopolistic Competition (Chapter 14)

- Monopolistic competition modeled as a four-stage game.
- Contrast with pure/perfect competition: in perfect competition, products are undifferentiated and firms are price takers; in monopolistic competition, firms differentiate their product (often through branding) and gain some pricing power as a result.
- Product differentiation examples discussed in class: restaurants (why a name/brand matters to a customer's perception of quality), national chains vs. independent local businesses, and family-owned businesses (an Italian-food restaurant was used as a running example).
- In-class problem: the goal is to maximize profit under a given monopolistic-competition scenario, then compare that outcome to what perfect competition would predict.

Assignment / in-class activity

- An exercise tied to the Chapter 14 monopolistic-competition rules — students work through a scenario and aim to maximize profit.
- Submission must include a screenshot embedded in the write-up to receive credit.
- A due date was mentioned but not clearly audible — confirm the exact date on the course site / announcements page.

Administrative notes

- Live attendance is not mandatory; sessions are recorded and posted for anyone who misses (consistent with the standing course policy).

- The announcements area is the source of truth for weekly assignments — check near the bottom of the announcements for the current week's items.
- Discussion questions continue on the same weekly cadence as prior modules.
- Worksheets are provided along with guidance on how to fill them out.
- For personal or individual issues, email the instructor directly rather than raising them in the general course channel.

See the Module 1 Class Meeting Summary in this folder for course-wide logistics (15-minute rule, exam format, etc.) established earlier in the term.